FloodLens AI: Multimodal Earth Observation & Disaster Intelligence Platform
Operational Rapid Flood Damage Assessment & Settlement Isolation Modeling

Technical Paper & Systems Report
Multimodal AI Hackathon 2026 — Track B: Mapping Flood Damage from Space
Study Focus: Trishuli River Corridor, Nuwakot District, Nepal (Monsoon Surge Event, August 2026)

Abstract
Rapid emergency response following monsoonal flood events in high-relief mountainous topography requires rapid, reliable, a

This paper introduces **FloodLens AI**, an open-source multimodal geospatial intelligence platform that pairs Sentinel-1 C-

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1. Problem Statement & Motivation
During extreme monsoon rainfall events in the Himalayan river valleys of Nepal—such as the Trishuli River basin—flash flooc

1. **Optical Cloud Blindness**: Persistent monsoonal cloud decks render optical sensors (Sentinel-2, Landsat) ineffective f
2. **Radar Topographic Cast Shadows**: In steep canyon valleys, terrain slopes cast deep radar shadows that exhibit low bac
3. **Target Leakage in Baseline Infrastructure**: Post-event crowdsourced mapping edits on OpenStreetMap inadvertently alte
4. **LLM Hallucination Risks in Emergency Reporting**: Commercial Large Language Models (LLMs) applied to disaster reports

FloodLens AI addresses these challenges through a deterministic geospatial pipeline combined with strict guardrail-groundec

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2. System Architecture & Multimodal Methodology

The FloodLens AI architecture comprises four decoupled operational layers operating across a FastAPI asynchronous backend a

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Coordinate Reference Systems (CRS)
To ensure zero area distortion in Himalayan topography, all spatial calculations operate across two standardized reference
- **Projected CRS — EPSG:32645 (UTM Zone 45N)**: All metric areas (m^2, km^2), segment lengths (km), buffer radii ($3
- **Geographic CRS — EPSG:4326 (WGS 84)**: All GeoJSON vectors, raster GeoTIFF georeferencing, and Leaflet interactive map

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3. SAR Inundation Extraction & Topographic Masking

The SAR processing pipeline processes dual-polarized ($VV+VH$) Sentinel-1 Interferometric Wide (IW) Ground Range Detected (

3.1 Radiometric Calibration & Log-Ratio Change Detection
Raw digital numbers (DN) are calibrated to radar backscatter cross-section σ^0 in decibels (dB):
$$\sigma^0\ (\text{dB}) = 10 \cdot \log_{10}(DN^2 + \epsilon)$$
Bi-temporal change detection computes the logarithmic amplitude difference between post-event (σ^0_{post}) ar
$$\Delta\sigma^0\ (\text{dB}) = \sigma^0_{\text{post}} - \sigma^0_{\text{pre}}$$
Candidate flood pixels are identified where the backscatter drops significantly below the empirical threshold $\tau_{\text{

3.2 Granular Radar Speckle Suppression
A spatial 2D median filter with an odd kernel (3×3 pixels) is applied to suppress high-frequency speckle noise while

3.3 Digital Elevation Model (DEM) Slope Masking
In high-relief terrain, mountain ridges facing away from the radar look direction produce dark cast shadows. To eliminate f
$$\theta = \arctan\left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\
All pixels where terrain slope $\theta > 15.0^\circ$ are unconditionally masked as mountain terrain shadow (`CLASS_TERRAIN_

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4. Infrastructure Exposure & Road Connectivity Analysis

4.1 Pre-Event Baseline Pinned Extraction
To prevent target leakage from post-disaster edits, infrastructure vectors are pinned to the pre-event historical snapshot

Critical healthcare facilities (hospitals, health posts, clinics) are classified into three proximity categories:
1. `DIRECTLY_INTERSECTING`: Ground footprint directly overlaps candidate flood polygons (0.0 m).
2. `BUFFER_ZONE_30M`: Structure centroid is located within 30.0 m of the flood boundary.
3. `SAFE_ELEVATED`: Structure is beyond the 30 m hazard buffer.

4.2 Topological Road Network Graph Modeling
The road network is modeled as a weighted multigraph $G = (V, E, W)$ using NetworkX in projected UTM Zone 45N:
- **Edge Weights**: Segment traversal times are assigned based on functional road classes: Primary Trishuli Highway H02 ($5
- **Settlement Snapping**: Settlement centroids are snapped to the nearest road network vertex within a configurable thresh
- **Simulated Road Closures**: Edges that intersect candidate flood polygons or inundated bridge coordinates are severed fr
- **Dijkstra Shortest-Path Routing**: Shortest-path distances and travel times from each community to the district referral

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5. Grounded AI Situation Report Copilot & Safeguards

The situation report engine translates structured numerical outputs into actionable, concise bilingual reports in **English

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[Structured Pipeline Metrics]
(Flood km2, Exposed Buildings, Severed Routes)
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|  Deterministic Bilingual Template Engine |
|  - Injects exact verified numbers        |
|  - Translates to Nepali Devanagari       |
|  - Enforces 'UNAVAILABLE' for missing    |
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|       Safety & Grounding Verifier       |
|  - Regex pattern scan: deaths, casualties|
|  - Blocks ungrounded evacuation orders   |
|  - Falls back if validation fails        |
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|       Multi-Format Document Exporter     |
|  - UTF-8 Markdown                        |
|  - ReportLab PDF (Nirmala.ttc font)      |
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5.1 Verification Guardrails
Before displaying any generated report, the engine applies automated grounding verification:
- **Prohibited Casualty Fabrications**: Scans for patterns matching `\b(death|deaths|casualty|casualties|fatalities|injured
- **Unauthorized Evacuation Orders**: Rejects authoritative language (`mandatory evacuation`, `official government order`).
- **Zero-Value Preservation**: Missing upstream layers report `UNAVAILABLE` rather than substituting zero-valued metrics.

5.2 Multi-Format Export
Reports are exported directly as UTF-8 Markdown and publication-grade PDF documents. For Nepali PDF generation, the system

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6. Benchmark Results & Verification

Testing was conducted using the benchmark case study **`e2e-trishuli-aug2026`** across the Trishuli River Corridor (`[85.05

Evaluation Metric	Analysis Value	Status
Analyzed Footprint	436.61 km^2	Full AOI coverage
Candidate Flood Extent	4.245 km^2	Synthetic SAR test fixture
Permanent Water Mask	18.192 km^2	Normal Trishuli riverbed
DEM Slope Mask ($>15^\circ$)	400.677 km^2	Terrain shadow suppressed
Potentially Exposed Buildings	3 footprints ($30,298.15\text{ m}^2$)	Pinned historical baseline
Potentially Inundated Roads	1.165 km	Segment intersections
Compromised Bridges	1 structure	Battar Suspension Footbridge
Intersecting Health Facilities	1 clinic	Battar Community Health Post
Newly Disconnected Settlements	2 communities	Devighat, Battar (severed to Bidur)
Baseline Isolated Settlements	1 community	Tupche (pre-event isolated)
Unmapped Settlements	1 community	Salme (reported as `UNAVAILABLE`)

All 96 backend unit and end-to-end integration tests pass, and the frontend production bundle builds cleanly.

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7. Known Limitations & Real-World Validation Requirements

FloodLens AI is currently evaluated as a **Methodological & Educational Prototype**. The following real-world data limitati

1. **Synthetic Radar Data**: Real Sentinel-1 GRD archives have not been downloaded and processed locally; all flood masks r
2. **Synthetic Infrastructure Baseline**: Historical Overpass Attic API queries returned HTTP 406; pre-event infrastructure
3. **Simulated Disconnections**: Road closures represent topological graph cuts, not confirmed physical pavement washouts c
4. **Field Ground-Truthing**: Satellite backscatter thresholding cannot distinguish between deep turbulent floods and shall

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8. Conclusion
FloodLens AI demonstrates that combining Sentinel-1 SAR change detection with pinned historical OpenStreetMap topologies ar

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Mandatory Data Attributions & Disclaimers

- **Copernicus Sentinel Data**: Contains modified Copernicus Sentinel data [2026].

- **Copernicus Digital Elevation Model**: Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence a
- **OpenStreetMap Data**: © OpenStreetMap contributors, licensed under the Open Database License (ODbL).
- **Educational Prototype Notice**: This system is a technology demonstration for the Multimodal AI Hackathon 2026. Outputs